INCH-POUND
MIL-D-24304B(SH)
AMENDMENT 3
7 December 1999
SUPERSEDING
AMENDMENT 2
22 February 1991

MILITARY SPECIFICATION

DIFFERENTIAL PRESSURE TRANSDUCER EQUIPMENT (ELECTRICAL) (NAVAL SHIPBOARD USE)

Inactive for new design after 21 August 1995.
For new design use MIL-T-24742.

This amendment forms a part of MIL-D-24304B(SH), dated 5 July 1988,and is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.1: Delete designations "D" and "SG" in their entirety and change "ST" to "GP."

PAGE 2

1.2.1.2: Delete in its entirety.

1.2.1.3: Delete in its entirety.

1.2.1.4: Delete and substitute:

"1.2.1.4 Application. The fluid pressure to be measured is indicated as follows:

GP - Steam, oil, fresh water, condensate, and other gases
SW - Seawater
FG - Flue gas and ammonia
OX - Oxygen"

1.2.1.7: Delete and substitute:

"1.2.1.7 Pressure rating. The pressure rating is designated by its numerical value. The minimum pressure rating is 150 lb/in^2 and not less than 300 percent of the range (see 6.2.1)."

AMSC N/A FSC 6685

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PAGE 3

2.1.1, under SPECIFICATIONS, FEDERAL: Delete "PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes."

2.1.1, under SPECIFICATIONS, MILITARY: Delete "MIL-E-17555 – Electronic and Electronic Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.; MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.; and MIL-C-52211 - Components and Assemblies for Industrial Gas Production, Storage and Transport Equipment, Packaging of."

PAGE 4

2.1.1, under STANDARDS, MILITARY: Delete "MIL-STD-275 - Printed Wiring for electronic Equipment, MIL-STD-278 - Welding and Casting Standard.; and MIL-STD-454 - Standard General Requirements for Electronic Equipment."

2.1.2: Delete and substitute:

"2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSEA 803-5959187	- Pressure/Differential Pressure Transducer Equipment Application/Selection/Installation Guide.
NAVSHIPS 810-1385846	- Oxygen System, Submarine Replenishment.

(Application for copies should be addressed to: Commander, Portsmouth Naval Shipyard, Code 202.2, Portsmouth, NH 03801.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIPS 0900-LP-001-7000	- Fabrication and Inspection of Brazed Piping Systems.
NAVSEA S9074-AR-GIB-010/278	- Requirements for Fabrication Welding and Inspection and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels.

(Application for copies should be addressed to the Defense Automated Printing Service (DAPS), 700 Robbins Avenue, BLDG. 4D, Philadelphia, PA 19111-5094.)"

PAGE 5

3.4.1: End the sentence after "24423" by adding a "." and delete the remaining portion ", and shall conform to requirement 4 of MIL-STD-454 for fungus inertness."

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PAGE 6

TABLE I: Delete and substitute:

"TABLE I. Material.

Sensing element and wetted parts	Steam, oil, fresh water, condensate, and other gases	Seawater	Service fluid flue gas and ammonia	Oxygen
CRES 304L, 316L, and 321	X	---	X	---
CRES 17-4 PH 17-7 PH 15-5 PH	X	---	---	---
Monel and K-Monel	X	X	---	X
Inconel 600 and X750	X	---	---	---
Inconel 625 and 718	X	X	---	---
Hastelloy C276	X	X	X	---
Titanium CP and 6A1-4V	X	X	---	---
CuNi 70/30	X	X	---	X
Ni span	X	---	---	---
Tantalum	X	X	---	---

"

PAGE 7

TABLE II: Delete and substitute:

"TABLE II. Ranges.

Type PDT	
Differential pressure ranges (PSID)	Differential pressure water column ranges (inches WCD)
0-15 0-30 0-60 0-100 0-200 0-400 0-600	0-10 0-30 0-60 0-150 0-300

PAGE 8

3.5.5 (f), line 2: Delete "2.0" and substitute "1.5".

PAGE 9

3.5.6, after last sentence: Add "The electrical receptacle may be designed to include built-in radio frequency suppression elements."

3.5.7, line 7: Delete "MIL-STD-278" and substitute "NAVSEA S9074-AR-GIB-010/278."

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3.5.9, lines 4 and 5: Delete "Oil may be present only in detectors for SW, OS, and FG applications."

PAGE 10

3.5.9 (a), line 2: Delete "packaged in accordance with MIL-C-52211" and substitute "pressure piping connections capped."

3.5.9 (b), line 1: Delete "GW" and substitute "GP, SW, and FG".

3.5.6 (c): Delete in its entirety.

3.5.9 (d): Delete in its entirety.

3.5.9.1, line 1: Delete "GW" and substitute "GP".

3.5.10: Delete and substitute:

"3.5.10 Welding and brazing. Internal pressure containing parts shall be joined by welding or brazing. Welding shall be in accordance with NAVSEA S9074-AR-GIB-010/278 and brazing in accordance with NAVSHIPS 0900-LP-001-7000.

3.5.10.1 General applications. Joints shall be either welded or nicrobrazed. Depending on the pressure range, NAVSEA S9074-AR-GIB-010/278, class P-1 or P-2 shall apply for welding. NAVSHIPS 0900-LP-001-7000 Class P-3 shall apply for nicrobrazing.

3.5.10.2 Oxygen applications. Joints shall be welded for all pressure ranges. NAVSEA S9074-AR-GIB-010/278, class P-1 shall apply."

3.5.12, line 2: Delete "GW" and substitute "GP".

Add as new paragraph:

"3.5.14 Label plate. A metallic label plate with engraved or stamped markings shall be permanently affixed to the outside of each differential pressure transducer. The label plate shall not be placed on the underside of the mounting surface or on any portion, such as an access cover, which may be separated from the body of the differential pressure transducer. As a minimum, the label plate shall contain the following information:

(a) Differential pressure transducer
(b) Manufacturers name and CAGE code
(c) National Stock Number
(d) Contract number
(e) Classification designation (see 1.2.1)
(f) Differential pressure range"

3.6.5, second line: Delete "MIL-STD-275 and".

PAGE 13

3.7.23, third line: Delete ":CE01, CE03, CS01, CS02, CS06, CS09, RE01, RS01, RS02, RS03," and insert ", table II,".

3.7.23 (a): Delete "CS02" and substitute "CS114".

3.7.23 (a)(1), first line: Add "megahertz" after "2".

3.7.23 (a)(2): Delete "shown in the CS02 test block diagram of" and substitute "as specified in".

3.7.23 (a)(4): Delete in its entirety.

3.7.23 (b): Delete "RS03" and substitute "RS103".

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3.7.23 (c): Delete in its entirety.

3.7.23 (d): Delete "CE01 and CE02" and substitute "CE101 and CE102".

3.7.23 (d) (1): Delete and substitute:

"3.7.23 (d) (1) The emission signal shall be measured only on the a.c. power leads of the test sample."

3.8: Delete in its entirety.

Add as new paragraph 3.8:

"3.8 Technical data. See appendix."

PAGE 14

4.1.2: Delete in its entirety.

4.3, after last sentence, add: "Except for electromagnetic interference emission and susceptibility, all examinations and tests shall be conducted utilizing a single transducer, one for each range (see 4.3.1), and in the order listed in table III. The electromagnetic interference emission and susceptibility test may be conducted on a separate transducer of the same design and differential pressure range. In addition, this test may be conducted concurrently with any of the other examinations and tests listed in table III. Deviations from the above procedure require the prior approval of NAVSEA."

PAGE 16

4.4, lines 4 and 5: Delete "be cause for rejection." and substitute "result in rejection of that transducer."

PAGE 17

4.5, line 4: Delete "no transducers shall" and substitute "that lot of transducers shall not".

4.5, line 10: Delete "results of all repeated tests" and substitute "based on test results".

4.7.2: Delete in its entirety.

PAGE 18

4.8.3, line 5: Delete "0.25" and substitute "1.00"; Line 10: Delete "0.25" and substitute "1.00".

4.8.13, second sentence: Delete "with a maximum pressure of 10,000 lb/in^2". Delete the last sentence.

PAGE 22

4.8.17: Delete and substitute:

"4.8.17 Differential pressure whip. Before performing the differential pressure whip test, a reference measurement shall be made (see 4.8.2). The test shall be conducted on a system capable of applying a periodic differential pressure change of not more than 20 percent to not less than 80 percent of transducer span for a total of 260,000 cycles. The rate of cycling shall be 1.0 Hz + 0.5 Hz. Additionally, the time for that portion of the cycle with differential pressures greater than the average of the maximum and minimum excursion points shall be between 33 percent and 67 percent of the time for one

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complete cycle. The transducer shall be energized throughout the test. After completion of the differential pressure whip test, a reference measurement shall be made for comparison (see 4.8.2). Performance shall conform to 3.7.16."

4.8.18, line 2: Delete "500" and substitute "50".

PAGE 23

4.9: Delete in its entirety.

5. PACKAGING: Delete and substitute:

"5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity."

PAGE 24

6.2.1(g) through (k): Delete and substitute:

"(g) National Stock Number (NSN), when available."

PAGE 25

6.2.2 and 6.2.2.1: Delete in their entirety.

LAST PAGE

At the end of document, add attached appendix, Recommended Engineering Drawings Technical Content Requirements.

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"APPENDIX A

RECOMMENDED ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

A.1. SCOPE

A.1.1 Scope. This appendix contains the format and content preparation instructions for the development of the revision to a conformance verification drawing. It is not intended that each requirement contained herein should be applied to every type of instrumentation. Portions of this appendix are subject to deletion tailoring depending upon the material, construction and principle of operating requirements that are specified in the individual instrumentation specification or acquisition document. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

A.2. APPLICABLE DOCUMENTS

A.2.1 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
ANSI Y14.1 - Drawing Sheet Size and Format.
ANSI/IEEE 200 - Standard Reference Designations for Electrical and Electronics Parts and Equipment.

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036.)

A.3. CONFORMANCE VERIFICATION DRAWING FOR INSTRUMENTATION

A.3.1 Purpose. The conformance verification drawing contains the information necessary to verify that the instrumentation meets the requirements specified in the applicable instrumentation specification and acquisition document.

A.3.2 Content and format. One conformance verification drawing should be developed for each specific type of instrumentation. It should include all ranges, sizes, connections and other variations. The conformance verification drawing should include the following minimum information (except as specified in A.1) and should be developed to the following format:

A.3.3 Descriptive data.

(a) Instrument identification numbering system.
(b) Test approval data.
(c) Conformance verification drawing acceptance data.

A.3.4 Details of construction.

(a) Two or more representative assembly views.
(b) Dimensions required to ensure interchangeability.
(c) Description of the nature and purpose of any adjustments.
(d) Complete weight.
(e) Location, size, and type of connections.

A.3.5 Electrical system schematics. Electrical schematics shall be included in the verification conformance drawing when applicable. A single schematic representing clearly the operation and the function of the electrical circuitry within the instrumentation. The schematic should contain all parts (including components, connectors, alarms, etc.) which make up the electrical

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system.

A.3.6 Selection and installation considerations.

(a) Performance data.
(b) Dimensional outline of the instrumentation showing overall and principle dimensions in sufficient detail to establish space requirements in all directions necessary for installation.
(c) Special considerations which may affect selection or installation.

A.3.7 Drawing format. The contractor's standard drawing format should be used.

A.4. CONFORMANCE VERIFICATION DRAWING ACCEPTANCE

A.4.1 Acceptance. Acceptance will be granted by the authorized Government activity only after the conformance verification drawing is found to meet all the requirements specified in A.2 through A.3.2.

A.5. Acquisition document instructions. The acquisition document should contain provisions that address submission, review, extension, disapproval, default, acceptance, and waiver of conformance verification drawings in addition to the effects on the delivery schedule due to delays in conformance verification drawing acceptance. NAVSEA 05Q is designated as the activity that accepts or disapproves the conformance verification drawing."

Custodian:
Navy - SH

Reviewer activity:
DLA - GS

Preparing activity:
Navy - SH
(Project 6685-0037)

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